**St. Francis Institute of Technology, Mumbai-400 103**
**Department Of Information Technology**

**A.Y. 2025-2026**
**Class: BE-ITA/B, Semester: VIII**
Subject: BlockChain Lab

# Experiment – 6

1. **Aim:** Case study on Hyperledger blockchain.
2. **Lab outcome:** After performing the experiment, the students will be able to **comprehend** the working of Hyperledger blockchain. (PO1, PSO1, BL2)

3. **Prerequisite:**
   Fundamental knowledge of blockchain
4. **Requirements:** The following are the requirements –
   PC and Internet.

5. **Pre-Experiment Theory:**
   What is Hyperledger?
   Hyperledger is an open-source project created to support the development of blockchain-based distributed ledgers. Hyperledger consists of a collaborative effort to create the needed frameworks, standards, tools and libraries to build blockchains and related applications. Hyperledger was created by the Linux Foundation in 2016.
   Hyperledger Fabric, an open-source project from the Linux Foundation, is the modular blockchain framework and standard for enterprise blockchain platforms. Intended as a foundation for developing enterprise-grade applications and industry solutions, the open, modular architecture uses plug-and-play components to accommodate a wide range of use cases.
   Hyperledger Composer is a set of collaboration tools for building blockchain business networks that make it simple and fast for business owners and developers to create smart contracts and blockchain applications to solve business problems. Built with JavaScript, leveraging modern tools including node.js, npm, CLI and popular editors, Composer offers business centric abstractions as well as sample apps with easy to test devops processes to create robust blockchain solutions that drive alignment across business requirements with technical development.
   A smart contract in Hyperledger Fabric is a program, called chaincode which is written in Go, JavaScript, and other programming languages such as Java that implement a prescribed interface.

6. **Laboratory Exercise**
7. **Steps to be implemented.**
   Select any one research paper regarding case study on Hyperledger Blockchain, study it and write report based on your studies.
8. **Program Code**
   Write any one sample chaincode in Golag for hyperledger

9. **Post Experimental Exercise-**
10. **Questions:**
    How a chaincode is implemented on Hyperledger blckchain?
11. **Results/Observations/Program output:**
    Present the program input/output results if any and comment on the same.
12. **Conclusion:**

a. Write what was performed in the experiment.
b. Write which tools you used to perform the experiment
c. Write what you inferred from the output obtained.

**13. References:**


[1] https://www.hyperledger.org/learn/case-studies

[2] https://hyperledger-fabric.readthedocs.io/en/release-2.2/whatis.h

[3] Hyperledger Fabric In-Depth: Learn, Build and Deploy Blockchain applications Using Hyperledger Fabric, Ashwani Kumar, BPB publications

Name: Vishal R Mahajan Experiment No: 06

Class: BE INFT B Roll No: 1

**Select any one research paper regarding case study on Hyperledger Blockchain, study it and write a report based on your studies.**

**Title:** Blockchain based Service: A Case Study on IBM Blockchain Services and Hyperledger Fabric

**Authors:** Bhuvana R. and P. S. Aithal

**Institution:** Srinivas University, Mangalore, India

**Published in:** International Journal of Case Studies in Business, IT, and Education (IJCSBE)

**Link:**
https://srinivaspublication.com/wp-content/uploads/2020/05/9.IBMBlock_FullPaper.pdf

## 1. INTRODUCTION

The rapid digital transformation in India has necessitated the shift from centralized data management to decentralized, secure, and transparent systems. This research paper explores how the combination of IBM's infrastructure and the Hyperledger Fabric framework provides a robust "Blockchain as a Service" (BaaS) model for enterprises. Unlike public blockchains that prioritize anonymity, this case study focuses on "Permissioned Blockchains" where participants are known and vetted, making it ideal for the Indian corporate and legal environment.

## 2. CORE TECHNOLOGY: HYPERLEDGER FABRIC

Hyperledger Fabric is identified as the backbone of enterprise blockchain solutions. The paper highlights several unique characteristics that distinguish it from other platforms:

- **Modular Architecture:** It allows components such as consensus and membership services to be "plug and play."
- **Permissioned Network:** Participants must be authorized to join, ensuring that sensitive business data is not accessible to the general public.
- **Channels for Privacy:** This feature allows for private communications between specific members of the network. For example, a supplier and a buyer can have a private channel for pricing that is not visible to other competitors on the same network.
- **Chaincode:** This is the platform's version of Smart Contracts, written in standard programming languages like Go, Java, or Node.js.

Name: Vishal R Mahajan Experiment No: 06

Class: BE INFT B Roll No: 1

## 3. THE ROLE OF IBM BLOCKCHAIN SERVICES

IBM acts as a facilitator by providing the "Cloud Environment" required to host these decentralized networks. The research details how IBM Blockchain Platform simplifies the deployment:

- **Ease of Use:** It provides a graphical interface to manage peers, orderers, and certificates.
- **Interoperability:** It allows organizations to connect their local data centers to the IBM Cloud.
- **Scalability:** Businesses can start with a small pilot and scale to thousands of transactions per second as the network grows.

## 4. ABCD ANALYSIS OF THE CASE STUDY

The authors employ the ABCD framework (Advantages, Benefits, Constraints, and Disadvantages) to evaluate the effectiveness of this technology in the Indian context:

**Advantages**

- High security through cryptographic hashing.
- Immutable records that cannot be altered or deleted.
- Elimination of middle men in complex transactions.

**Benefits**

- Real time tracking of goods in the supply chain.
- Significant reduction in administrative costs.
- Enhanced trust between business partners who may not know each other.

**Constraints**

- Significant initial investment in cloud infrastructure.
- A shortage of skilled blockchain developers in the current market.
- Regulatory uncertainty regarding digital ledger technology in some sectors.

**Disadvantages**

- High complexity in the initial setup phase.
- Potential performance bottlenecks if the network is not optimized correctly.

## 5. INDUSTRIAL APPLICATIONS IN INDIA

The paper discusses how this specific case study applies to various Indian sectors:

- **Banking and Finance:** Streamlining Cross Border payments and reducing the time taken for "Know Your Customer" (KYC) verification.
- **Supply Chain:** Tracking the movement of agricultural products from rural farms to urban markets to ensure freshness and fair pricing.
- **Healthcare:** Securing electronic medical records so they can be shared between hospitals without compromising patient privacy.

## 6. RESEARCH FINDINGS

The study concludes that the integration of Hyperledger Fabric with IBM services provides a "Single Version of Truth" for all stakeholders. The researchers found that businesses using this model reported:

1. Increase in operational efficiency.
2. Improved transparency in auditing and compliance.
3. Enhanced data security compared to traditional SQL based centralized databases.

## 7. CONCLUSION

The research by Bhuvana and Aithal serves as a comprehensive guide for Indian enterprises looking to adopt blockchain. By utilizing a permissioned framework like Hyperledger Fabric, organizations can achieve the benefits of decentralization while maintaining the privacy and control required for commercial operations. The "BaaS" model is projected to be the primary driver of blockchain adoption in India over the coming decade.

Name: Vishal R Mahajan Experiment No: 06

Class: BE INFT B Roll No: 1

```
package main
import (
	"fmt"
)
type Asset struct {
	ID    string
	Name  string
	Value string
}
var ledger = make(map[string]Asset)
func CreateAsset(id string, name string, value string) {
	asset := Asset{
		ID:    id,
		Name:  name,
		Value: value,
	}
	ledger[id] = asset
	fmt.Println("Asset Created Successfully")
}
func ReadAsset(id string) {
	asset, exists := ledger[id]
	if !exists {
		fmt.Println("Asset does not exist")
		return
	}
	fmt.Println("Asset Details:")
	fmt.Println("ID:", asset.ID)
	fmt.Println("Name:", asset.Name)
	fmt.Println("Value:", asset.Value)
}
func main() {
	CreateAsset("1", "Laptop", "50000")
	ReadAsset("1")
}
```

```
Asset Created Successfully
Asset Details:
ID: 1
Name: Laptop
Value: 221068

Program exited.
```